

UBT510: BIOLOGY FOR ENGINEERS

L	T	P	Cr
2	0	0	2.0

Course Objective: To learn about living world and basic functioning of biological systems. The course encompasses understanding of origin of life and some of its central characteristics. It also aims to familiarize engineering students to some of the intricate biological phenomena and mechanisms.

Syllabus

Introduction to the living world: Origin of life and chemical evolution, Properties of Life, levels of biological organization, Cells as unit of life, structure and functions of cell organelles.

Biomolecules and their applications: Brief introduction to the molecular constituents (Protein, carbohydrates, Lipids and nucleic acids) of living cells and their salient structural & functional attributes. DNA as information storage devices, DNA computing, DNA origami in synthesizing 2D and 3D structures, DNA fingerprinting, molecular scissors.

Energy and life: Law of energy transformation, free energy and metabolism, ATP generation, role of enzymes in metabolic reactions.

Nature inspired Biomaterial: Biomechanics, Biomimetics, Nanostructures in living world and their extraordinary properties, gecko foot, lotus leaf, turkey egg shell, mother of pearl, special properties of spider web and silk fibers, biosensors, smart implants, bionics, biosensors, medical diagnostics based on molecular recognition.

Computational aspects of bioinformatics: Sequence alignment, scoring matrices, algorithms used in sequence alignment, multiple sequence alignment and its applications.

Laboratory Work (if applicable)

Course Learning Objectives (CLO)

The students will be able to:

1. Explain the characteristic features of living-systems and differentiate them from non-living systems
2. Broaden the application of engineering knowledge of their branch by applying concepts of living systems.
3. Demonstrate familiarity with special properties of biological macromolecules
4. Upgrade their understanding about biological systems by drawing parallel with thermodynamics system and develop interface between an engineering specialization and living systems.

Approved in 1st meeting of the Academic Council held on September 11, 2025. Further revision approved in 3rd meeting of the Academic Council held on March 18, 2026, incorporating modification in credits due to the addition of AI courses.

5. Design engineering products inspired by living creatures.
6. Plan application of computational tools in bioinformatics.

Text Books

1. Nelson, DL and Cox MM., Lehninger: Principles of Biochemistry, WH Freeman (2008) 5th ed.
2. Campbell, NA, Reece, JB, Urry LA, Cain, ML, Wasserman, SA, Minorsky, PV and Jackson, RB, Biology – A global approach, Pearson Education Limited (10th edition).
3. Ghosh Z, and Mallick B, Bioinformatics – Principles and Applications, Oxford University Press (2008).

Reference Books

1. Mount D W, Bioinformatics - Sequence and Genome Analysis, Cold Spring Harbour Laboratory Press (2001).
2. Bruce Alberts et al., Essential cell biology, Garland Science (Taylor & Francis Group).

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1.	MST	25-35
2.	EST	35-45
3.	Sessionals (May include assignments/quizzes)	30

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